

Diabetes Detection Using Supervised Learning

Executive Project Summary

Problem: Predict whether a patient has diabetes using clinical measurements.

Approach: Performed data cleaning, EDA, feature engineering, scaling, and compared multiple supervised learning models using cross-validation.

Algorithms Used: Logistic Regression, Random Forest, Decision Tree, and KNN.

EDA Findings: Glucose, BMI and Age showed the strongest relationship with diabetes. Insulin had 374 missing values and SkinThickness had 227 missing values after treating invalid zero values as missing.

Feature Engineering: Replaced invalid zero values with missing values, filled missing values using median imputation, standardised features, and evaluated reduced feature sets.

Model Comparison: Best model: Logistic Regression (without Insulin & SkinThickness) Accuracy 77.08%, ROC-AUC 83.80%. Next best model: Logistic Regression (without Insulin) Accuracy 76.95%, ROC-AUC 83.79%. Random Forest achieved Accuracy 76.68%, ROC-AUC 83.46%.

Why This Model?: Insulin and SkinThickness contained the largest amount of missing data. Removing them reduced noise while slightly improving Accuracy, Recall, F1-score, PR-AUC and ROC-AUC. The model also remains simple, interpretable and easier to deploy.

Final Result: Selected Model: Logistic Regression without Insulin and SkinThickness. Accuracy 77.08%, Precision 72.16%, Recall 57.11%, F1 Score 63.48%, ROC-AUC 83.80%, PR-AUC 72.90%.

Model	Accuracy	ROC-AUC
Logistic Reg (-Insulin,-SkinThickness)	77.08%	83.80%
Logistic Reg (-Insulin)	76.95%	83.79%
Random Forest (All Features)	76.82%	83.50%